



Unit 4 · Outcome 3 · Scientific Investigation

From a column of numbers to evidence

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



00 Where this sits

KK07 is the analysis engine of your whole investigation. Whatever you measured in KK03, this dot point asks: what does it actually show, and how sure can you be?

Turn raw readings into a form you can read — clean tables, repeat trials averaged, processed columns, a sensible graph.

Pull the pattern out — the trend, the relationship between variables, the line or curve of best fit, the size of the effect.

Judge how trustworthy it is — anomalies, sources of error, uncertainty, and whether the data really supports the claim.

◆ KEY INSIGHT

Key Students who only describe the graph ("it goes down") top out around half marks. The evaluate band — error, uncertainty, anomalies — is where 12+ students separate from 11s.

RAW READINGS → ORGANISE → ANALYSE → EVALUATE → JUSTIFIED CONCLUSION

01 The big picture

Raw data is gravel and gold mixed together. Analysis is the panning.

◆ ANALOGY

Analogy · panning for gold A pan of river gravel is your raw data — full of value, but you can't see it yet. Organising is washing out the obvious mud (tidy tables, averaged repeats). Analysing is the swirl that separates the heavy gold from the light sand — the trend rising out of the noise. Evaluating is checking your gold is real and not fool's gold: is that glint a genuine pattern, or a flake of error?

Every step has a matching exam skill, and they run in this order. Skip organising and your graph is unreadable. Skip evaluating and your conclusion is overconfident. The diagram below is the spine of this whole module — each stage links to a section.

02 The investigation we'll analyse

One dataset carries the whole module, so every technique lands on the same real numbers.

Priya, a Year 12 student, noticed Wirrim Creek runs cloudy below a stormwater outfall after rain. She designs an investigation:

Question — Does turbidity (water cloudiness) change with distance downstream from the stormwater outfall?

Controlled — same probe, same depth (10 cm), measured within one hour, 36 h after rain

Method — at each of 8 sites she takes 3 repeat readings and records them in her logbook.

• NOTE

Why turbidity NTU = Nephelometric Turbidity Units. Higher NTU = cloudier water = more suspended sediment, often washed in by stormwater.

03 Organising the data

Before any graph: clean the table, average the repeats, decide your processed column.

Three readings per site is raw data. The mean of those three is processed data — it's the value you graph. Averaging repeats is the single most important organising move: it smooths random scatter and gives you one fair number per site.

Distance (m)	Trial 1 (NTU)	Trial 2 (NTU)	Trial 3 (NTU)	Mean (NTU)
0	82	88	85	85.0
25	71	69	73	71.0
50	58	55	61	58.0
75	44	47	42	44.3
100	36	95	34	35.0*
125	27	25	29	27.0
150	19	21	20	20.0
175	15	13	14	14.0

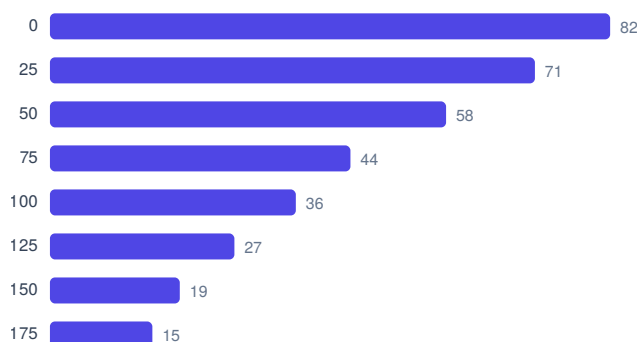


Figure · Distance (m) compared at a glance.

* The 95 NTU reading is anomalous — we'll deal with it in §06. The mean shown excludes it (mean of 36 and 34), which Priya justifies in her logbook.

✓ TIP

Organising checklist Headers with units · IV in the left column · consistent decimal places (sig figs) · repeats then a mean · anomalies flagged, not deleted.

⚠ WATCH OUT

Exam trap Don't "tidy" raw data by silently dropping a number you dislike. Every excluded value needs a logged reason. Examiners check for honesty here.

04 Finding the trend

A graph isn't decoration — it's the analysis. The fit line is your claim about the relationship.

Plot IV on the x-axis, DV on the y-axis, one point per processed value. Then add a line or curve of best fit — a single smooth line that passes through the centre of the trend, balancing points above and below. It is not a connect-the-dots zig-zag.

⚠ WATCH OUT

Trap · join-the-dots "Join the dots" implies every wiggle between points is real. It isn't — those wiggles are mostly random error. A best-fit line says this is the underlying trend.

• NOTE

Line vs curve Use a straight line only if the data is genuinely linear. Priya's data curves — turbidity falls fast then levels off — so a curve of best fit models it far more honestly than a straight line.

◆ ANALOGY

Analogy · the line of best fit is a compromise candidate Think of each data point as a voter pulling the line toward itself. The line of best fit is the candidate who can't please everyone but sits closest to the whole electorate — equally disappointing above and below. Connect-the-dots is letting every single voter veto the path; you end up with chaos, not a policy.

05 Patterns & relationships

Naming the relationship precisely is worth marks. "It goes down" is not the language of an A.

Negative here: as distance $\uparrow$, turbidity $\downarrow$. A positive relationship rises together; no relationship shows no consistent trend.

Non-linear: a steep fall that flattens toward a baseline (~ 12 NTU). Not a straight line — the rate of change shrinks with distance.

Strong: points sit tightly on the curve with little scatter, so the trend is clear and the data is reliable.

⚠ WATCH OUT

Trap · correlation $\neq$ causation Priya's graph shows turbidity correlates with distance. To claim the outfall causes it, she needs her controlled variables to rule out other explanations (a second drain, a tributary, cattle access). A pattern is evidence for a cause, never proof of one on its own.

Interpolation — reading a value between measured points (e.g. turbidity at 60 m) is reliable; the line supports it.

Extrapolation — extending the line beyond your data (e.g. at 300 m) is risky; you have no evidence the trend continues. State which one you're doing.

06 Outliers & anomalies

An anomaly is a reading that doesn't fit the pattern. Spotting it, investigating it, and justifying what you do with it is a classic mark-grabber.

At 100 m, trial 2 read 95 NTU while its repeats read 36 and 34. It sits wildly above the trend — a clear anomaly. The right process is never "delete and move on":

⚠ WATCH OUT

Trap · anomaly $\neq$ uncertainty An anomaly is a single rogue reading from a one-off cause. Uncertainty is the small spread present in every measurement. Don't call ordinary scatter an "anomaly" — examiners notice.

07 Sources of error

Two families, two fixes. Confusing them is the most common error-question mistake.

Scatters readings both ways around the true value — sediment clouds drifting past, slight probe-depth differences, ripples. Affects precision (the spread).

Reduce it by repeating and averaging. More repeats = smaller random error in the mean.

Shifts every reading the same way — a probe calibrated against a wrong standard reads, say, 4 NTU high every time. Affects accuracy (closeness to truth).

Repeating won't fix it. You catch it by calibrating against a known standard and checking your method.

◆ KEY INSIGHT

The one-liner examiners want Random error → fix by repeating & averaging. Systematic error → fix by calibrating / correcting the method. Naming the right fix for the right error type is the mark.

08 Uncertainty

No measurement is a single perfect number. Quantifying the doubt is what makes data scientific.

The spread of Priya's repeats is her uncertainty. At 0 m she read 82, 88, 85 — a range of 6 NTU. A simple, VCE-acceptable estimate is $\pm$ half the range, so 85 ± 3 NTU. On a graph, that becomes an error bar — a small whisker through each point showing how far the value could reasonably swing.

If two points' error bars overlap, you can't confidently say they differ. If they're clearly separated, the difference is real. Error bars turn "looks different" into a defensible claim.

Reliable = repeats agree (small error bars). Valid = you measured what you meant to, with confounders controlled. Data can be reliable but invalid — perfectly consistent and consistently wrong, if a systematic error is baked in.

✓ TIP

Tip Toggle error bars back on the §04 graph. Notice the bars are small where repeats were tight and larger near the anomaly site — a visual map of where Priya can trust her data most.

09 Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you exactly how much analysis to show.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

10 Named exam traps

⚠ EXAM TRAPS

Each of these has cost real students real marks. Learn them by name.

⚠ WATCH OUT

Trap 1 · the bald description "The graph goes down" earns almost nothing. Quote values and name the shape: "turbidity falls from 85 to 14 NTU, steeply at first then levelling off."

⚠ WATCH OUT

Trap 2 · join-the-dots A zig-zag line through every point treats random error as real signal. Draw one smooth line/curve of best fit.

⚠ WATCH OUT

Trap 3 · wrong fix for the error "Repeat readings to reduce systematic error" — wrong. Repeats fix random error. Systematic needs calibration.

⚠ WATCH OUT

Trap 4 · deleting anomalies silently Removing the 95 NTU value with no logged reason looks like fudging. Flag it, give the cause, justify exclusion.

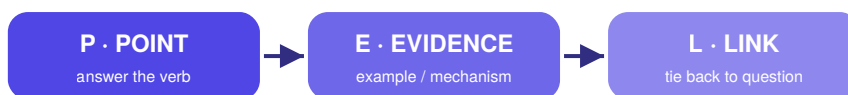
⚠ WATCH OUT

Trap 5 · correlation = causation "Distance causes turbidity" overclaims. The data shows a relationship; controlled variables let you argue cause.

⚠ WATCH OUT

Trap 6 · reckless extrapolation Predicting turbidity at 500 m from data ending at 175 m. You have no evidence the trend holds out there.

11 P.E.L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Analysis answers follow Point · Evidence · Link. Type below and watch the meters respond to your keywords.

✓ TIP

Point: name the relationship (negative / non-linear / strong). Evidence: quote numbers + units. Link: connect back to the hypothesis or mechanism.

12 Worked answers · weak vs strong

Same question, two responses. The difference is precision, not length.

Q. Using Table 1, describe the relationship between distance downstream and turbidity, and evaluate the reliability of the data. (4 marks)

The turbidity goes down as you go further away. The results are pretty reliable because she did the experiment carefully and got good numbers.

There is a strong negative, non-linear relationship: turbidity falls from 85 NTU at the outfall to 14 NTU at 175 m, dropping steeply then levelling toward a baseline. The data is largely reliable — the three repeats at each site agree closely (e.g. 82, 88, 85 at 0 m), giving small uncertainty of about ± 3 NTU. The one exception is the 95 NTU anomaly at 100 m, which disagrees with its repeats and was excluded with a logged cause, so it does not undermine the overall trend.

13 Practice exam

★ PRACTICE & SELF-MARK

Five graded questions · 20 marks. Reveal the guide, mark yourself honestly, and the dock tracks your total. A turbidity probe was calibrated against the wrong standard and reads 4 NTU too high at every site. This is an example of: Describe the relationship between distance downstream and turbidity shown in Priya's data. (Quote data in your answer.) ① Direction: negative — as distance increases, turbidity decreases (1). ② Shape: non-linear — steep fall then levelling off toward a baseline (1). ③ Quotes data with units: e.g. from 85 NTU at 0 m to 14 NTU at 175 m (1). Priya's repeats at most sites differ by only 2–6 NTU, but the probe was later found to read consistently 4 NTU high. Evaluate the impact of random and systematic error on her results, and state how each could be addressed. ① Random error is small — tight repeats (2–6 NTU) show good precision; reduced further by repeating & averaging (1). ② Systematic error of +4 NTU shifts every reading up equally, harming accuracy (1). ③ Repeats do not fix systematic error (1). ④ Fix: re-calibrate against a known standard / subtract the 4 NTU offset (1). In a different study, Jordan measures soil moisture (%) at increasing distance from a leaking irrigation pipe and gets: 0 m → 38%, 5 m → 31%, 10 m → 47%, 15 m → 19%, 20 m → 12%. Analyse the data: identify the trend, flag any anomaly with a justified decision, and evaluate one limitation of drawing a conclusion from this dataset. ① Trend: negative — moisture decreases with distance from the pipe (1). ② Quotes the fall, e.g. 38% → 12% (1). ③ Identifies 10 m → 47% as an anomaly breaking the trend (1). ④ Justified decision: investigate cause (e.g. a puddle / second water source); flag and likely exclude with a logged reason — do not silently delete (1). ⑤ Limitation: only one reading per distance — no repeats, so reliability can't be judged and the anomaly can't be confirmed (1). ⑥ Sensible extension: take repeats / add control sites away from the pipe to establish causation (1). "Priya's data proves the stormwater outfall causes high turbidity in Wirrim Creek." Evaluate this claim, referring to the pattern in her data, the difference between correlation and causation, and the role of her controlled variables. ① The data shows a strong negative relationship (turbidity highest at the outfall, 85 NTU, falling downstream) — quoted (1). ② This is a correlation: distance and turbidity vary together (1). ③ Correlation alone does not prove causation (1). ④ Her controlled variables (same probe/depth, fixed time after rain) strengthen the causal case by removing confounders (1). ⑤ But uncontrolled factors remain possible — a tributary, second drain, or bank disturbance (1). ⑥ Balanced conclusion: the data strongly supports but does not prove the outfall as cause; "proves" overstates it (1).

14 Recap · the analysis spine

◆ KEY INSIGHT

If you remember one thing The marks above 50% live in the evaluate band. Always close an analysis answer with error, uncertainty, and how much the data can really be trusted.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

The turbidity goes down as you go further away. The results are pretty reliable because she did the experiment carefully and got good numbers. No data quoted, no shape named — just "goes down." "Did it carefully" is not evidence of reliability; reliability is about repeats agreeing, which isn't mentioned. The 95 NTU anomaly is ignored entirely. $\approx 1/4$.

✓ STRONG / MODEL ANSWER

There is a strong negative, non-linear relationship: turbidity falls from 85 NTU at the outfall to 14 NTU at 175 m, dropping steeply then levelling toward a baseline. The data is largely reliable — the three repeats at each site agree closely (e.g. 82, 88, 85 at 0 m), giving small uncertainty of about ± 3 NTU. The one exception is the 95 NTU anomaly at 100 m, which disagrees with its repeats and was excluded with a logged cause, so it does not undermine the overall trend. Names direction + shape + strength, quotes specific values with units, defines reliability correctly via repeat agreement, quantifies uncertainty, and handles the anomaly transparently. $4/4$.

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